

Roll No.

TCS-711

B. TECH. (ML/DS/CCIS)

(SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

SPARK FUNDAMENTALS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Rephrase Apache Spark and MapReduce by taking suitable real time example. 10 Marks (CO1, CO6)

OR

- (b) Develop a code in Spark, for checking the frequency of each word present in a text document. Comment for each line of code is mandatory. 10 Marks (CO1, CO6)

2. (a) Showcase a few key features of Apache Spark. 10 Marks (CO1, CO2)

OR

- (b) By standards "Apache Spark is said to be a unified computing engine." Showcase the validity of this statement, i.e. whether it is true in case of Apache Spark. If yes, why? 10 Marks (CO1, CO2)

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(2)

3. (a) Identify the important components of Spark Ecosystem, helping businesses to boost their ROIs. 10 Marks (CO3, CO4)

OR

- (b) Interpret the execution of structured API. 10 Marks (CO3, CO4)
4. (a) Outline how Spark runs applications with the help of its architecture. 10 Marks (CO2, CO5)

OR

- (b) Identify the reasons for Spark being good at low latency workloads like graph processing and machine learning. Support your answer with the Spark APIs involved. 10 Marks (CO2, CO5)
5. (a) Relate various cluster managers available in Apache Spark. 10 Marks (CO5)

OR

- (b) "Partitioning is the core for any distributed framework, so as in case of Apache Spark." Enlighten this statement by your thought and examples. 10 Marks (CO5)